KIMBERLY (KIMI) YASHAR

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EDUCATION

Carnegie Mellon University - Pittsburgh, PA — Expected May 2030
- Bachelor of Science, Intended majors in Physics and Robotics
- Coursework: Fundamentals of Programming and Computer Science (15-112); Planned Spring 2027: Principles of Imperative Computation (15-122), Concepts of Mathematics (21-127), Introduction to Electrical and Computer Engineering (18-100)

The Harker School - San Jose, CA — Graduated May 2026
- Coursework: Neural Networks, Differential Equations, Multivariable Calculus, Data Structures

SKILLS

Programming & Scientific Computing: Python, Java, VPython, Mathematica
Hardware & Design: Raspberry Pi, Fusion 360 (CAD)
Laboratory: Scanning electron microscopy (SEM), optical microscopy, chemical etching, soldering

ENGINEERING & SOFTWARE EXPERIENCE

Software Team Member, Tartan Autonomous Underwater Vehicles — August 2026 - Present
- Developing and benchmarking a real-time ROS 2 RGB-D camera pipeline for an autonomous underwater vehicle using a Linux-based NVIDIA Jetson AGX Orin embedded system and Luxonis OAK-D camera.
- Evaluating MJPEG, H.264, and H.265 for RGB video and lossless PNG compression for 16-bit depth images; profiling bandwidth, frame rate, latency, CPU utilization, and image fidelity.

Raspberry Pi Photo Booth [GitHub] — June 2026 - August 2026
- Built a standalone Raspberry Pi 4B photo booth in Python with a camera, 7-inch touchscreen, live preview, and automatic photo storage.
- Integrated Bluetooth Low Energy printing with an Instax Mini Link printer and debugged GPIO connection failures to stabilize unattended operation.

Personal Engineering Invention — September 2022 - November 2023
- Designed and prototyped a low-friction hairbrush using porous bristles to generate air bearings; filed a U.S. patent application and submitted responses to two USPTO Office Actions [Patent].

RESEARCH & LAB EXPERIENCE

Student Researcher, SLAC National Accelerator Laboratory — July 2026 - August 2026
Neilander Group
- Fabricated antireflective silicon surfaces through anisotropic wet etching of single-crystal wafers in 2 M KOH and characterized the resulting pyramidal morphology using SEM.
- Constructed TiO_2/p-Si heterojunction solar cells and conducted electrolysis experiments documenting corrosion across zinc, tin, aluminum, and copper electrodes.

Materials Science Researcher & Microscopy Lead — May 2025 - May 2026
Starostina Lab, Aspiring Scholars Directed Research Program
- Coauthored “Chemical Etchants for Dihedral Angle Measurements of Twin Boundaries in Pure Copper” [Paper], accepted to the IEEE MIT Undergraduate Research Technology Conference.
- Presented findings to audiences totaling 300+ and designed and taught a 24-student course in microscopy, optical interferometry, and CAD.

LEADERSHIP

Founder, Discovery Day [Website] — February 2025 - Present
- Founded a hands-on science program to counter science department budget cuts, reaching 500+ students (30% English second language) thus far and growing the program to 6 directors.

Founder, Community Tables Cooking Classes [Gallery] — August 2024 - April 2026
- Founded and led monthly bilingual cooking classes for 300+ families at an East San Jose food-assistance nonprofit, securing $800 in grant funding.

HONORS & AWARDS

- 1st Place, Bay Builders Hackathon at AWS - AI-powered A/B testing platform - July 2026
- National Merit Finalist - Top 1% nationwide - February 2026
- Elite Grand Champion, Walk to End Alzheimer’s - Raised $28,000 over six years - 2026
- STEM Merit Scholarship, University of Cambridge - Top 5% of applicants - October 2024